500451

Course Name: Computer Networks

CO1: Understand the concept of OSI and TCP/IP reference model
CO2: Understand the basics of data transmission at physical layer.
CO3: Understand the channel allocation using ALOHA, CSMA and CSMA/CD.
CO4: Apply error detection and correction technique to eliminate transmission error.
CO5: Analyze the fixed and variable length address (IPv4) subnetting for the given scenarios.
CO6: Understand the design issues of the transport layer.
CO7: Understand the mechanism of protocols at application layer such as FTP, HTTP, Telnet, DNS.
CO8: Understand IPv6 addressing and differentiate it from IPv4.

Printed Pages:3

University Roll No. .....

End Term Examination, Even Semester 2024-25
B. Tech. (CSE/AIML&IIoT/BCT/CCV/DA/CSF), II Year, IV Semester
BCSC0008 & Computer Networks

Time: 3 Hours

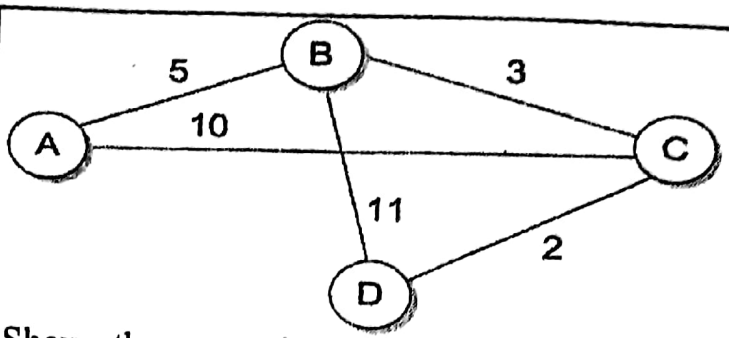
Maximum Marks: 50

Instruction for students: Attempt all questions

Section – A

Attempt All Questions 4 X 5 = 20 Marks

No.	Detail of Question	Marks	CO	BL	KL
1	Identify seven layers of the OSI model and explain their basic functions?	4	1	U	F
2	Explain the importance of CRC. A bit stream 110101101 is transmitted using the standard CRC method. The generator polynomial is 101. What is the actual bit string transmitted?	1+3	3	A	C
3	For each of the following networks, discuss the consequences if a connection fails. a) Five devices arranged in a mesh topology b) Five devices arranged in a bus topology	2+2	2	R	F
4	Given an IP packet of 4404 bytes (with a 20-byte header) and a router Maximum Transmission Unit (MTU) of 1500 bytes, what values should be set for the MF bit, Fragment Offset, and Total Length fields in the third fragment?	4	5	An	M
5	Consider the network shown below.	4	5	A	C



Show the operation of Dijkstra's (Link State) algorithm for computing the least cost path from A to all destinations. Also, explicitly list all the shortest path routes from A to all destinations that are the result of the algorithm's computation.

Section – B

Attempt All Questions 3 X 5 = 15 Marks

No.	Detail of Question	Marks	CO	BL	KL
6	How does DNS support internet communication and what are its three primary domain divisions?	3	7	R	F
7	Discuss the role of headers in HTTP communication. Explain any two HTTP methods.	1+2	7	U	C
8	Explain the role of following methods at Application Layer: a) WWW b) FTP c) TELNET	1+1+1	7	U	F
9	Your router has the following IP address on Ethernet0: 172.16.2.1/23. Calculate the Network ID and valid host IDs on the LAN interface attached to the router?	1.5+1.5	5	An	P
10	The following is the dump of a TCP header in hexadecimal format: 06540017 23140001 221200000 500207FF 00000000 (a) What is the sequence number? (b) What is the destination port number? (c) What is the acknowledgment number?	1+1+1	6	A	P

Section – C

Attempt All Questions 5 X 3 = 15 Marks

No.	Detail of Question	Marks	CO	BL	KL
11	Explain the purpose of SYN, SYN-ACK and ACK in three-way handshake protocol used in TCP to establish a reliable connection between two hosts. Also, explain how sequence number and acknowledgment number are used.	5	6	A	P
12	Give the basic steps of RSA Algorithms. In a RSA cryptosystem, a participant A uses two prime numbers p=5 and q=7 to generate its public and private keys. If the public key of A is 5, then the private key of A is? For the plaintext value 2, show the process of cipher text generation using RSA cryptosystem.	2+1+2	8	A	P
13	A company is facing issues with its email system where employees can receive emails but are unable to send them. Upon inspecting the mail server logs, the administrator notices that outgoing emails are not being relayed properly via the mail server. a) Explain the function of SMTP in the email delivery process and why it is crucial for sending outgoing emails. b) Explain the functions of POP and IMAP in the email delivery process.	3+2	7	C	M